

Metadata-Augmented Recursive Decomposition for Cost-Efficient Comprehension of Unbounded Documents

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Abstract—Recursive Language Models (RLMs) process inputs far beyond a model’s native context window by holding the document in a REPL environment and letting a root model explore it programmatically, delegating reading to sub-model calls. We observe that this exploration is *structurally blind*: each recursive call receives a raw text slice with no view of the document’s organisation, no record of what sibling calls have already established, and no statement of why it was invoked. On documents whose structure is exploitable — textbooks, manuals, standards — this discards precisely the information that makes the document navigable.

We propose Metadata-Augmented Recursive Decomposition (MARD), an inference-time extension that threads a *metadata envelope* through every recursive call. The envelope carries a structural skeleton derived from the document’s own text, findings accumulated from prior calls, and a parent-issued directive, and it grows as exploration proceeds. Exploration thereby becomes an act of confirming a hypothesis rather than of discovering blindly. We instantiate MARD on dependency-ordered study-material generation over textbook-scale documents, where a frontier “Scout” reads a small fraction of the document to emit a typed Master Plan and a fleet of cheap “Swarm” builders generate material that is joined in dependency order rather than book order.

[TODO: Results sentence. Blocked on results freeze A (27 Aug 2026). Report the O3 delta against vanilla RLM in both quality and tokens, and state the O4 boundary. Per docs/00-CLAIM.md the claim is a *disjunction* — quality and/or tokens — so a win on either is the claimed result and a null is reported as a null.]

Index Terms—long-context language models, recursive inference, document structure, prerequisite ordering, inference-time methods, cost-efficient inference

to a sub-model that reads it and returns an answer. The root is an orchestrator that decides *what to read* and delegates *the reading*. Because the root sees only truncated program output, its own context grows slowly, and inputs of several million tokens become tractable against a context window two orders of magnitude smaller.

A. The defect: recursion without orientation

We observe a specific deficiency in this design. Every sub-call is given a raw slice of text and nothing else. It does not know which document the slice comes from, where in that document it sits, what other sub-calls have already established, or why it in particular was invoked. It is, in effect, a reader handed three photocopied pages with no cover and no page numbers.

Two costs follow. First, orientation is re-derived on every call rather than established once and reused. Second, sub-calls cannot build on one another: a finding established in Chapter 3 is unavailable to the call that reads Chapter 9, even though the document itself may explicitly connect them. On a *flat* context — a shuffled transcript, a log file — neither cost is avoidable, because there is no structure to be oriented against. But on a document with exploitable structure, the information being discarded is exactly the information the author supplied to make the text navigable: a table of contents, numbered sections, and in-text cross-references of the form “as we saw in Chapter 3.”

B. Contribution

We introduce a *metadata envelope*: a compact, growing structure carried into every recursive call, comprising (i) a structural skeleton of the document, (ii) findings accumulated across prior calls, and (iii) a directive issued by the calling context stating what this call is for. Our claim, stated in the form we commit to before reporting any measurement:

MARD’s growing metadata envelope makes recursive document exploration structure-aware — improving output quality and/or reducing tokens consumed relative to vanilla RLM on documents with exploitable structure, while degrading gracefully to vanilla-RLM behaviour on documents without it.

I. INTRODUCTION

The dominant approach to reasoning over documents that exceed a language model’s context window is to reduce the document: chunk and retrieve, compress the prompt, or compact the conversation. Each discards information and each does so before the model has had a chance to decide what matters.

Recursive Language Models [1] invert this. The document is never placed in the prompt at all. It is bound to a variable inside a Python REPL, and the root model receives only *metadata* about it — its type, its length. The root then writes code to explore it: slicing, regular expressions, loops. Inside that REPL it may call `llm_query`, handing an extracted slice

A second property of the envelope emerged from measurement rather than design, and we report it as a finding rather than folding it into the claim above: because MARD’s plan is a *typed contract* validated at the tier boundary, a builder receives its source span by construction rather than by a model matching text to a concept. We observed the vanilla baseline silently generate content without consulting the document at all, while reporting success (Section V-B). The envelope’s structural apparatus makes that failure mode loud rather than silent, which is a claim about *faithfulness and auditability* distinct from output quality.

Three properties of this sentence are deliberate. It credits the envelope, not recursion, because recursion is the base paper’s. It is a *disjunction* over quality and tokens, because we do not know in advance which axis the effect appears on and measure both independently. And the graceful-degeneration clause is a prediction stated in advance, not a failure mode discovered afterwards; it is an objective of this work rather than a caveat on it.

C. What is not claimed

It is important to say this before a reader infers otherwise. The base paper’s RLM already pairs a frontier root model with cheaper sub-model calls, and already recurses. Neither is our contribution. The base library that accompanies it already carries rich per-call metadata and even ships an example demonstrating metadata captured at every depth of the recursion tree. Section III-A draws the line precisely: that metadata flows *upward* and is *observational*, whereas MARD’s envelope flows *downward* and is *operative*. Same word, opposite direction, and only one of the two changes what the model does.

We also make no claim to beat the state of the art on long-context benchmarks. Isolating MARD’s contribution means comparing against vanilla RLM; the base paper’s own Table 1 reports a higher overall high-water mark from a different system, and that remains the standing bar [1].

D. Demonstrating application

We instantiate MARD on the generation of dependency-ordered study material from textbook-scale documents. This application is chosen because it makes the envelope’s contribution measurable rather than anecdotal: a textbook declares its own prerequisite relations through in-text cross-references, so a generated ordering can be scored against ground truth extracted programmatically from the document itself, with no expert annotation and therefore no human-subjects component.

II. RELATED WORK

a) *Recursive and long-context inference.*: Our work extends RLM [1] directly. A recent critique, SRLM [2], reports gains over RLM and argues that recursion may not be the primary driver of RLM’s advantage, observing that within-window recursion sometimes degrades performance. We take no position on that dispute, and our design does not require one: the envelope reduces *orientation cost*, which is a

claim about what each call is given rather than about what makes recursion work. The critique does, however, shape our methodology. If the objection to a reported improvement is that it may be sampling noise or prompt luck, the defence is not more documents but variance across repeated runs, a negative control, and a boundary condition stated in advance. Section IV adopts all three.

Positional-degradation results [3] motivate the broader family of context-reduction approaches, of which prompt compression [4], [5], [6] and its recent survey [7] are representative. These are complementary rather than competing: they reduce what a call must read, whereas MARD changes what a call *knows* about what it reads.

b) *Hierarchical decomposition of long documents.*: Hierarchical merging for summarisation [8] and multi-agent narrative summarisation [9] decompose long inputs into a tree and combine partial results upward. The structural difference from MARD is the direction of information flow: in merging schemes, context is assembled on the way up. In MARD, accumulated context is injected on the way down, before each unit of work executes.

c) *Tiered and cost-aware inference.*: Routing and cascading approaches [10], [11] and hint-based delegation [12] allocate a frontier model selectively to control cost. MARD’s Scout/Swarm split is an instance of the same economics, and we claim no novelty for it. What is ours is that the cheap tier receives a *typed plan directive* rather than an undifferentiated slice — an ablation we isolate directly (Section V-D).

d) *Prerequisite structure and educational applications.*: Concept prerequisite extraction has an established literature [13], [14], [15], as do LLM agents in education [16], [17]. We differ in the source of supervision: rather than learning prerequisite relations from labelled corpora, we treat each edge our system emits as a *falsifiable prediction* against the document’s own declared cross-references.

e) *Positioning against shipped systems.*: Two honest acknowledgements, which we state rather than let a reader supply. Ingestion capacity is *not* the gap — deployed products already accept large document collections — and at least one shipped product now generates sequenced study plans from uploaded course material. The line we draw is on the *source* of the sequencing: deployed systems that sequence do so diagnostically, ordering by what a learner has answered incorrectly, whereas our ordering is document-derived, extracted without supervision and scored against the text’s own cross-references. We deliberately do not use any closed product as a quantitative baseline: their architectures are unpublished, they expose no token accounting, and they change without notice, so no such comparison would reproduce. **[VERIFY: CONTEXT.md Risk 2 — the hierarchy-vs-dependency distinction has no peer-reviewed support. The only admissible evidence is first-hand screenshots placed beside our own concept graph. Do not cite tufino2025notebook1m here; it is a physics-education paper and does not support this claim. Owner: Track 4 / Track 1.]**

III. METHOD

A. What the base library already provides

Formalising MARD requires first naming what it does *not* add, because the base library ships a great deal that sounds like our contribution. Its metadata types carry the root model, maximum depth, iteration count, backend and environment; its completion object carries the full execution trajectory — iterations, code blocks, and nested sub-calls, each with its own metadata, recursively; and per-call token and cost accounting is built in.

That metadata, however, flows *upward* and is *observational*: it is a record of what happened, assembled for inspection after the fact. It never re-enters a prompt and never influences a child call. This is checkable in the source rather than a matter of framing. When a parent spawns a child, the child is constructed with a fresh, empty logger — the library’s own comment says the purpose is that the child’s trajectory be *captured*, not that the parent’s be communicated — and the child receives only the prompt string that the parent’s REPL code passed to it. Nothing else crosses the boundary.

MARD’s envelope flows *downward* and is *operative*. Skeleton, accumulated findings and parent directive enter the child’s context *before* it runs, changing what the child can see and therefore what it produces.

One objection should be anticipated. The base library does expose a slot for injecting a small user-authored string alongside the root’s context, and declines to populate it for children. The rebuttal is about what fills the slot, not whether a slot exists: that parameter is designed for a fixed, short, user-supplied string — its documentation’s own example is the user’s original question. The envelope is accumulated state that grows across calls. No mechanism in the base library constructs, propagates or compounds such state, and none carries one child’s findings to the next. The library gives us somewhere to put an envelope; the envelope is ours.

B. The metadata envelope

Let D be a document and $\mathcal{S}(D)$ a *skeleton*: an ordered structural index of D ’s chapters and sections with their positional spans, derived from D ’s own text. For a recursive call c we define the envelope

$$E(c) = \langle \mathcal{S}(D), F(c), d(c) \rangle,$$

where $F(c)$ is the set of findings accumulated by calls ordered before c , and $d(c)$ is the directive issued by c ’s parent stating the purpose of the call. $E(c)$ is rendered into c ’s prompt in addition to its text slice. The envelope is bounded by construction: our per-call budget is approximately 300 tokens, against a design ceiling of 500. **[VERIFY: Confirm the 300-token figure against the rendered envelopes in corpus/introcs/pass1_trace.json before this goes to the guide. It is the per-chapter figure; section-level rendering measured an order of magnitude higher and is why Pass 1 dispatches per chapter.]**

The design constraint that makes this non-trivial is that $E(c)$ must remain small while $F(c)$ grows monotonically with the

number of completed calls. Findings are therefore summarised, not concatenated.

C. Pass structure

MARD proceeds in passes over D , all executed by the frontier tier.

- **Pass 0 — skeleton extraction.** Derive $\mathcal{S}(D)$ from the document’s text. This pass is predominantly arithmetic over detected block structure rather than model inference; a single labelling call assigns topic labels. Crucially, the skeleton is *text-derived only*: where a source document ships a publisher-supplied bookmark outline, that outline is used solely as a yardstick for evaluating skeleton quality and is never an input (Section IV-C).
- **Pass 1 — enriched exploration.** One call per chapter, each carrying $E(c)$ and each contributing to F . The output is the *Master Plan*.
- **Pass 2 — targeted deep dive.** Revisit spans that Pass 1 marked as uncertain, at greater depth.

a) *This manuscript reports a two-pass configuration.:*

The system evaluated here executes Pass 0 and Pass 1 only; Pass 2 is not run. This is stated here, in every results caption, and in the limitations, because describing a two-pass system as three-pass would be an overclaim. The decision was taken deliberately: Pass 1’s implementation was completed one day before this project’s feature freeze, and the reliability of a pass cannot be evidenced in a day. Layering a targeted deep dive on an unevidenced exploration pass would make a subsequent wrong result undiagnosable — it could not be attributed to either pass. The two-pass configuration is additionally, and by design, the depth-0 point of the depth sweep in Section V-D, so this manuscript’s number is a valid point on that curve rather than a one-off compromise.

D. The Master Plan as a typed contract

Pass 1 emits a *Master Plan*: a concept graph together with an ordered study sequence and the rationale for any reordering. It is validated against a schema at the tier boundary, and a malformed plan fails loudly rather than dispatching N subtly incorrect builders.

The field that carries the contribution is the concept graph’s edge set. Each edge asserts that one concept is prerequisite to another, and each is therefore a *falsifiable prediction* scorable against the document’s own in-text cross-references.

E. Two-tier execution and dependency-ordered joining

The **Scout** (frontier tier, single session) reads approximately 3–5% of D and emits the Master Plan. The **Swarm** (budget tier) then runs N builders in parallel, one per section, each receiving its section slice plus its plan directive. Because the builders fork and join, wall-clock time is $\max_i(\text{builder}_i)$ rather than $\sum_i$, and we report both so that the parallelism claim is checkable rather than asserted.

Builder outputs are joined in *Master Plan order* — the dependency-derived ordering — rather than in book order. We note explicitly that book order is itself an expert baseline,

TABLE I
CORPUS DOCUMENTS AND LICENCES, VERIFIED AGAINST EACH
SOURCE’S OWN COPYRIGHT PAGE.

Document	Licence
<i>Introduction to Computer Science</i> (OpenStax)	CC BY 4.0
<i>University Physics, Vol. 1</i> (OpenStax)	CC BY-NC-SA 4.0
<i>Linear Algebra Done Right</i> , 4th ed.	CC BY-NC 4.0

since the document’s author already sequenced the material; reordering must therefore beat a real ordering, not a random one.

F. The stated precondition

MARD requires exploitable structure. On a flat context there is no skeleton to extract, F carries no positional meaning, and the envelope reduces to overhead. We therefore predict that MARD degenerates to vanilla-RLM behaviour on flat inputs, and we test this with a negative control rather than omitting it. This is an objective of the work, not an embarrassment to be discovered in review.

IV. EXPERIMENTAL SETUP

A. Corpus and licensing

Our corpus consists of openly licensed textbooks spanning distinct domains, under an open-license-only constraint on all source material. Table I summarises them. Licensing was verified against each publisher’s own copyright page rather than a third-party aggregator.

OpenStax titles were selected specifically because they ship machine-readable learning objectives, glossaries and answer keys — the substrate required for document-native ground truth extracted without human annotation, and therefore without an ethics-approval process this project’s timeline cannot absorb.

a) *On the licence of the primary document.*: *Introduction to Computer Science* is CC BY 4.0 (plain attribution), confirmed against OpenStax’s book-level licensing page. Individual chapter footers within the same book state CC BY-NC-SA; we treat the book-level page as authoritative and record the inconsistency rather than silently choosing one.

[VERIFY: OSTEP contradiction. CONTEXT.md 2.1 lists OSTEP among the four corpus documents; the corpus rationale here excludes it as standard-copyright. docs/16-PRIMARY_DOCUMENT.md already records that the corpus is “effectively two usable documents plus one restricted one” against the planned four. This must be resolved before Manuscript B, which needs two further openly licensed candidates that pass the parse-quality checks below.]

B. Ingestion and primary-document selection

Primary-document selection was made on measured evidence rather than convenience. Table II reports parse quality across the three candidates. No optical character recognition is used on any document: all three carry a text layer, and OCR would replace exact text with approximate text —

TABLE II
PARSE QUALITY, PRODUCED BY THE INGESTION PIPELINE AND
REPRODUCIBLE FROM THE CORPUS ARTEFACTS. *Orphan punctuation*
COUNTS PUNCTUATION STRANDED WHERE A FORMULA FAILED TO
EXTRACT.

Role	intros	physics1	axler
Primary			
Secondary	939	959	404
Secondary			
Characters after cleaning	2,432,812	2,032,210	737,844
Headings detected	1,403	1,081	17
Learning-objective blocks	61	99	0
Key-terms blocks	14	17	0
Orphan punctuation (per 1k lines)	30.3	113.5	87.8
Warnings raised	0	1	3

decisively, OCR errors in glossary terms would corrupt the very reference set against which task quality is scored, silently and irreversibly.

University Physics is disqualified as primary on content loss, not formatting loss: its orphan-punctuation rate is nearly four times that of the primary document, and inspection confirms sentences from which equations have vanished entirely. That damage is not uniform across systems under test — a configuration that reads more of the document encounters more of it — and would manufacture a difference between MARD and vanilla RLM having nothing to do with either. *Linear Algebra Done Right* carries no learning objectives and no key-terms sections, so document-native ground truth cannot be extracted from it; it remains the strongest candidate for the dependency-ordering case study, which is scored separately.

Page mapping was verified by sampling 1,200 blocks at random across the three documents and confirming each appears on the page it claims; 1,199 passed, the single failure being a whitespace-normalisation difference in an index line rather than a mapping error.

C. Two structural confounds, and how they are controlled

Both concern structure reaching a model without that model having derived it. If structure arrives by another route, we would be reporting the publisher’s work as MARD’s.

- 1) **The publisher’s bookmark outline.** All three documents ship one. The ingestion pipeline writes it to a separate artefact with an explicit provenance field and never merges it into the text stream, so that using it must be a deliberate decision rather than an accident. It serves as the yardstick for skeleton quality and is not an input to Pass 0.
- 2) **The printed table of contents inside the body text.** Subtler, because it arrives through the ordinary text channel: the primary document prints its full chapter list on page 7, so a model reading the opening pages obtains the skeleton for free. Front matter is detected, tagged, and excluded from the text stream supplied to every system.

Both controls are applied identically to the vanilla-RLM arm. If the two arms received different inputs, the comparison

would not isolate the envelope. **[VERIFY: docs/16 verification debt item 2 records this as open — confirm Pass 0 and A1 are in fact treated identically in the harness, not merely intended to be.]**

D. Skeleton fidelity

Because the skeleton is the substrate for everything downstream, its quality is measured before it is used. Against the publisher outline, the primary document’s text-derived skeleton achieves **82.2% recall at 0.0 pages of boundary error**. Boundary error of zero indicates that where a section is recovered, its span is recovered exactly; the recall shortfall is missed sections, not misplaced ones.

E. Systems compared

- **MARD** — two-pass, as specified above.
- **Vanilla RLM** — the isolation baseline. Identical models, document, and controls; the envelope removed, so each recursive call receives its raw slice only. This is simultaneously the primary baseline and ablation A1, and is executed once, not twice.
- **Negative control** — a flat-context task, to test the graceful degeneration half of the claim.

Manuscript B additionally runs full-context, naive chunking and embedding-RAG baselines first-hand.

a) *Both arms are executed on a single implementation.*: Ablation A1 requires that the MARD and vanilla-RLM arms differ *only* in the presence of the envelope. Executing the baseline on a separate codebase would confound the envelope with two independent implementations — differing prompt templates, REPL surfaces and stopping criteria — none of which is the variable under study. We therefore implement vanilla RLM as an envelope-disabled configuration of the same system, and treat the reference implementation accompanying the base paper [1] as the specification we implement against rather than as the artefact we measure.

The cost of this choice is that our baseline is our own implementation of RLM rather than the original authors’. We mitigate it in the only way available: by reproducing a published result from the base paper first-hand on our own harness before any number in Section V is trusted. That check is not a formality here — it is the sole evidence that the vanilla arm faithfully reproduces the published method. **[TODO: Report the reproduced base-paper figure and the target it was checked against. This check gates every number in Section 5.]**

b) *Concurrency and sub-call size are bounded, identically for both arms.*: Runs were executed under a provider rate-limit tier that required bounding sub-call concurrency and sub-call input size. **[TODO: State the two values, k concurrent sub-calls and m tokens per sub-call, once fixed.]** Both bounds are applied to *both* arms and are recorded in every run’s configuration snapshot. We report this because it is not a neutral implementation detail: throttling can degrade output completeness, and applying it asymmetrically would manufacture a difference between the arms that has nothing

to do with the envelope. Where a bounded run produced a truncated artefact, the run is recorded as *failed* rather than scored as degraded — see the join-time completeness check below.

c) *Truncation is a failure, not a low score.*: A builder response can be correctly positioned, built for the correct concept, and still carry no text — a safety refusal, a token-limit finish with nothing emitted, or a tool-call-only completion. Sorting such a set succeeds quietly and yields an artefact short by one section with every identity check satisfied. The join therefore validates non-empty text per plan step and raises rather than returning a partial artefact, so a hole in the results matrix explains itself instead of appearing as a plausible low number.

d) *Recursion depth is held fixed.*: Both arms run at a matched recursion depth in which sub-calls are plain completions. This is the base paper’s primary reported condition. Holding depth fixed is not incidental: if the arms differed in depth, the measured difference would confound the envelope with an extra layer of recursion, and the isolation claim would be void. **[VERIFY: The mapping between MARD’s pass-depth numbering and the control library’s recursion-depth parameter is off by one, and is currently derived from project documentation rather than from the implementation’s call structure. Confirm against envelope/pass1.py before this paragraph is final — docs/18 5.2.]**

F. Metrics

Every run of every system records seven fields, and a number that cannot be traced to all seven from a logged run is not admitted to this paper: task score, tokens consumed (input and output, summed over every model call and reported separately in the log), calls issued split by tier, cost at the same-day published provider rate, wall-clock latency (both max and $\sum$ over builders), run identifier, and a full configuration snapshot including model identifiers, prompt-template versions, depth setting, active ablation, and document.

a) *Task score.*: The evaluated output modality is **explanations only**. Flashcards, quizzes and diagrams are demonstrable prototype features that test nothing about the envelope and would multiply the evaluation surface; they are excluded from measurement. Explanations are scored against document-native ground truth — glossary terms, per-chapter learning objectives, in-text cross-references and forward-reference positions — extracted programmatically. **[TODO: State the exact scoring function once Track 3’s scorer is final. The ground-truth source and the modality constraint are frozen; the function is not yet.]**

b) *Dependency-ordering score.*: Scored independently of task quality, as *forward-reference violations*: the number of times a generated sequence places a concept before something the document itself declares as its prerequisite, counted for book order and for Master Plan order.

c) *Variance.*: Every reported number is the result of three independent repeated runs, with spread reported alongside the

TABLE III

PROJECTED COST OF THE MANUSCRIPT A MATRIX ON THE PRIMARY DOCUMENT, FROM MEASURED CORPUS QUANTITIES. TOKEN COUNTS MARKED † ARE DERIVED FROM CHARACTER COUNTS AT AN ASSUMED RATIO AND ARE PENDING CONFIRMATION WITH A TOKENISER.

Measured quantity	Value
Primary document, characters	2,468,431
Primary document, tokens [†]	≈617,000
Sections (Swarm builders)	120
Chapters explored by Pass 1	14
Projected cost	
MARD, one run	—
Vanilla RLM, one run	—
Both arms, three repeats	—
Manuscript A matrix, complete	—

TABLE IV

MARD (TWO-PASS) AGAINST VANILLA RLM ON THE PRIMARY DOCUMENT. THREE REPEATED RUNS; SPREAD IN PARENTHESES. BOTH ARMS AT MATCHED RECURSION DEPTH WITH IDENTICAL STRUCTURAL CONTROLS.

System	Task score	Tokens	Cost
Vanilla RLM (= A1)	—	—	—
MARD (two-pass)	—	—	—
Δ	—	—	—

central value. We report repeats rather than fixed random seeds deliberately: the inference endpoints used here do not expose deterministic decoding, so a nominal seed would name a property the system does not have. Where variance swamps an effect, that is reported as the finding.

G. Implementation and cost

[TODO: Name the model pair and provider once selected, with same-day rates from the provider’s own pricing page and the date read. This is an open decision as of this draft — docs/18 7 item 1 — and blocks the cost column below.]

The cost model is instantiated with measured token counts rather than a claimed multiple. Table III reports the projected cost of the full Manuscript A matrix from measured corpus quantities. We report the model rather than a headline ratio because provider rates move month to month, and a fixed multiple quoted against a rate that later disappears is precisely the kind of claim that decays into misinformation.

V. RESULTS

[TODO: This entire section is scaffolding. No number below is real. Nothing may be written here that does not trace to a logged run under runs/. After the results freeze, numbers are written up and never re-run; a wrong result receives a limitation paragraph, not a second attempt.]

TABLE V

GROUNDNESS BY SYSTEM AND REPEAT. RATE IS GROUNDNESS CONCEPTS AS A FRACTION OF CONCEPTS IN THE FINAL ARTEFACT.

System	Repeat	Concepts	Grounded	Regen.
Vanilla RLM	1	—	—	—
Vanilla RLM	2	—	—	—
Vanilla RLM	3	—	—	—
MARD	1	—	—	—
MARD	2	—	—	—
MARD	3	—	—	—

A. Isolating the envelope

B. Groundedness of generated content

The most consequential observation in this study was not a score. In one repeat of the baseline, the system completed without error, reported success, and produced a coherent study guide in which a substantial fraction of the explanations had been generated *without the document in context at all*.

The mechanism is worth stating precisely, because it is a property of the method rather than of our harness. In vanilla RLM the root model authors its own exploration code. In this run that code mismapped a concept onto an unrelated passage; the resulting generation failed to parse and was discarded; and the retry proceeded with an empty source field, producing an explanation from the model’s parametric knowledge. Nothing in the returned artefact, the exit status, or the aggregate token accounting distinguishes that explanation from a grounded one. The failure is visible only by walking the execution trajectory.

We therefore instrument groundedness directly. For each concept in a final artefact we resolve the generating call and classify it as *grounded* (non-empty source text), *ungrounded* (empty or absent source), or *regenerated* (at least one prior attempt discarded). [TODO: Report the groundedness rate per run for both arms. Measure MARD’s rate and report it whatever it is — claiming immunity without measuring is the same overclaim inverted. State plainly that this was found by manual trajectory inspection and instrumented afterwards.]

MARD’s architecture bears on this structurally rather than incidentally. Builder inputs are constructed from the plan’s recorded source span rather than matched at generation time; the plan is schema-validated at the tier boundary and a malformed plan fails loudly rather than dispatching many subtly incorrect builders; and the join validates non-empty output per plan step, raising rather than returning an artefact that is short by a section while satisfying every identity check.

C. Variance across repeats

[TODO: Report the three baseline repeats individually. Do not average them: the spread in concept count, cost and wall-clock is itself the result. State that no repeat was excluded.]

TABLE VI
ABLATION A2 — PLAN WITHHELD FROM THE SWARM.

Configuration	Task score	Tokens
MARD (full)	-	-
A2 (plan withheld)	-	-

D. Ablation

The frozen ablation set comprises four manipulations: **A1**, the envelope removed, which is architecturally vanilla RLM and is therefore the same run as the baseline; **A2**, the plan withheld from the Swarm, in which the Master Plan is still computed and still determines join order but builders receive no directive; **A3**, reordering disabled, joining in book order; and **A4**, a sweep over recursion depth reported as a curve rather than a table row. This manuscript reports A1 and A2; A3 and A4 appear in the full study.

A2 is the ablation reported here because it separates two effects that A1 collapses together: whether *exploration* benefits from structure-awareness, and whether *dispatch* does.

[TODO: A2 failure mode to check before reporting: if a builder’s prompt template reconstructs plan-like context from the raw section text — because the section carries its own headings — this ablation under-measures the plan’s contribution. Log the exact prompt sent to each builder in this condition so the check is auditable rather than asserted.]

E. Negative control

[TODO: Flat-context result. The claim predicts degeneration to vanilla-RLM behaviour here. A null difference is the predicted outcome, not a failure.]

VI. DISCUSSION

A. The structure-dependence boundary

[TODO: Written against real numbers. The frame is fixed in advance: MARD is predicted to help where structure is exploitable and to degenerate where it is not. Both halves are results.]

B. Structure as a faithfulness mechanism

[TODO: Written against Table V. The argument: a typed, validated plan does not merely order the output, it constrains what each generation step is permitted to see. Where the baseline can fall back to parametric knowledge silently, the ablated pipeline either supplies the recorded span or fails. If MARD’s own groundedness rate is below one, report and explain that instead.]

C. Threats to validity

a) *The comparison rests on one mechanism.*: If the envelope ceases to propagate downward, MARD is vanilla RLM and every number in Table IV measures nothing — while every component still executes and no test fails. This mechanism is therefore guarded by dedicated tests asserting

that a later chapter’s prompt contains an earlier chapter’s findings, and that a child envelope carries its parent’s findings.

b) *Structure leakage.*: Section IV-C describes the two routes by which document structure could reach a model without being derived. We control both and apply the controls to both arms. We cannot rule out a third route we have not thought of.

c) *No repeat was excluded.*: One baseline repeat exhibited the grounding failure described in Section V-B, and produced markedly different output volume, cost and latency from the other two. It is reported. A run is treated as re-runnable only when the *protocol* failed to execute — a harness fault, a provider fault, an unverified corpus. A run in which the system under test behaved badly is a result. Excluding the repeat in which the baseline performed worst would be indistinguishable from tuning toward a positive outcome.

d) *Single primary document.*: Manuscript-scale evidence from one document cannot establish that an effect is not document-specific. This is why variance across repeats, a negative control, and a pre-stated boundary carry the argument here rather than breadth, and why the full study expands the corpus.

e) *Provider dependence.*: Results are obtained against a hosted inference endpoint whose decoding is not deterministic and whose behaviour may change. We mitigate by reporting repeats with spread and by recording a full configuration snapshot per run.

D. Limitations

The system evaluated here is **two-pass**: the targeted deep-dive pass is specified but not executed, for the reasons in Section 3. Evaluated output is restricted to explanations.

Our vanilla-RLM baseline is our own implementation of the published method rather than the original authors’ code, adopted so that both arms share one implementation and the comparison isolates the envelope (Section IV). Runs are additionally executed under bounded sub-call concurrency and input size imposed by our provider rate-limit tier; these bounds apply to both arms, but they place the absolute scores below what an unthrottled configuration would produce, and we therefore treat the *difference* between arms as the reportable quantity and the absolute values as configuration-specific.

Whether improved structural ordering translates into learning gain is not addressed — that requires a controlled study with human participants, which we scope as future work and do not speculate about here.

VII. CONCLUSION

[TODO: Draft last, after Section 5. One paragraph. State the mechanism, the measured direction of effect, and the boundary. Do not restate the abstract.]

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APPENDIX

This appendix is a working artefact and is deleted before submission. It exists so that no open item is lost between drafts.

- 1) **Model pair and provider unselected.** Blocks the cost column, the implementation paragraph, and the harness sanity check.
- 2) **Citation gate.** Only two of the seventeen bibliography entries have a verified row in the project’s literature log. Section 2’s prose is written so that no *claim* rests on an unverified entry, but every entry must be verified against its primary record before submission. Two author lists are explicitly unconfirmed.

- 3) **Title conflict** on the SRLM reference: two different titles are recorded for the same arXiv identifier. Resolve against the abstract page.
- 4) **The hierarchy-versus-dependency claim** has no peer-reviewed support. First-hand screenshots are the only admissible evidence.
- 5) **OSTEP** appears in the planned corpus but is excluded by the licensing constraint. The full study needs two further openly licensed documents that pass the parse-quality checks.
- 6) **Depth-numbering off-by-one** between the pass structure and the control library’s parameter, currently derived from documentation rather than code.
- 7) **Envelope token budget** quoted at approximately 300 tokens; confirm against the logged Pass 1 traces.
- 8) **Task-scoring function** not yet final.
- 9) **Harness sanity check** — reproduce a published base-paper number first-hand before any result in Section 5 is trusted. **Now load-bearing**, not routine: it is the only evidence that our own implementation of the baseline is faithful to the published method.
- 10) **Throttle values** k and m unstated, and their effect on absolute scores unquantified. Confirm both arms carry them.
- 11) **Structural-control parity** between the MARD and vanilla arms is intended but not yet confirmed in the harness.

[TODO: Master Plan schema; full ablation grid; the pinned commit of the control library, recorded by the exact revision measured; corpus artefact checksums; and the run-identifier scheme. Note the licence of the control library correctly — it is MIT.]